

Recap:

$$\underline{MT} \rightarrow U_1 \dots U_n, \overset{iid}{\sim} U(0,1)$$

$\Rightarrow$

$$\int f(x) dx$$

$$(1) \Rightarrow \underline{PIT} \rightarrow X \sim F, \text{ (conts)}$$

$$U \sim U(0,1)$$

$$X = F^{-1}(U) \rightarrow$$

limitation:

$$\left[\begin{array}{l} U \sim U(0,1) \\ X = F^{-1}(U) \sim \text{PRNG from dist}^n \\ F \end{array} \right]$$

(2) Acceptance Rejection method:

When the Inverse Transform Method Struggles

- Requires F^{-1} in **closed form** – not always available.
- **Classic failure case: the Normal distribution.** Its CDF $\Phi(x)$ has no elementary closed-form inverse.
 - Workaround 1: numerical approximation of Φ^{-1} (e.g. rational approximations, Newton–Raphson root finding on $\Phi(x) - u = 0$).
 - Workaround 2: use a specialized transformation method instead → **Box–Muller** (coming up).
- For multivariate distributions, F^{-1} is rarely tractable at all.
- Even when F^{-1} exists, it can be numerically unstable near $U \approx 0$ or $U \approx 1$ (extreme tails).

Motivates the next methods

When F^{-1} is unavailable or expensive, we turn to **acceptance–rejection**, **special transformations** (Box–Muller), and the **ratio-of-uniforms** method.

Outline

Pseudo-Random Number Generators

Generating Uniform and Non-Uniform Random Variables

Inverse Transform Method

Acceptance–Rejection Method

Setup and Theory

Proof of Correctness and Efficiency

Worked Examples

Box–Muller Transformation

Ratio-of-Uniforms Method

Diagnosing Randomness

Simulation Error

Unit 6

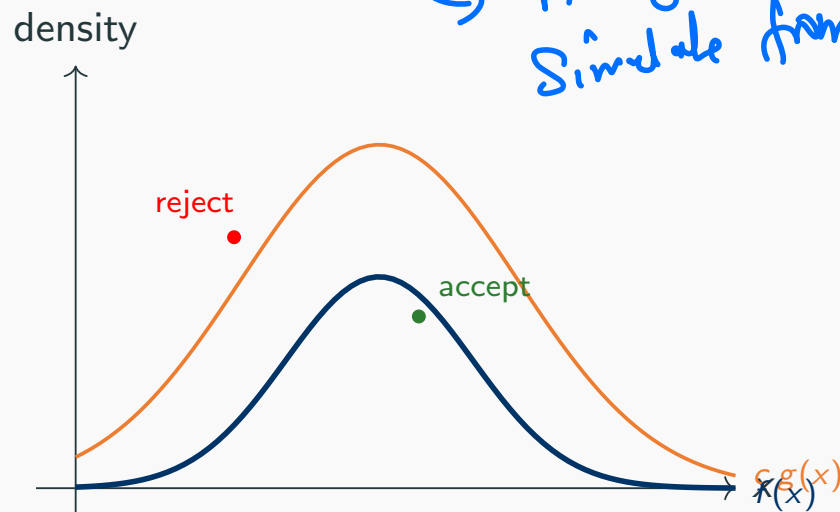
The Idea: Darts Under a Curve

$$E(g(x)) = \int g(x) f(x) dx \rightarrow \rightarrow X \sim f(\cdot)$$

$\rightarrow$ Pdf

- Target density $f(x)$, hard to sample directly.

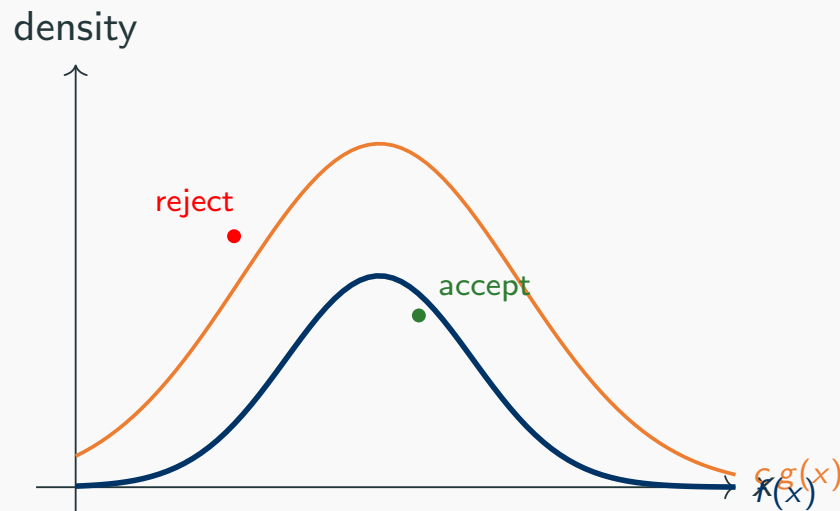
how you can simulate from f .



The Idea: Darts Under a Curve

- Target density $f(x)$, hard to sample directly.
- Choose an envelope/proposal density $g(x)$ that is easy to sample from.

f : Normal
 $\rightarrow g$: Cauchy $\rightarrow$ inverse
transformation
method.

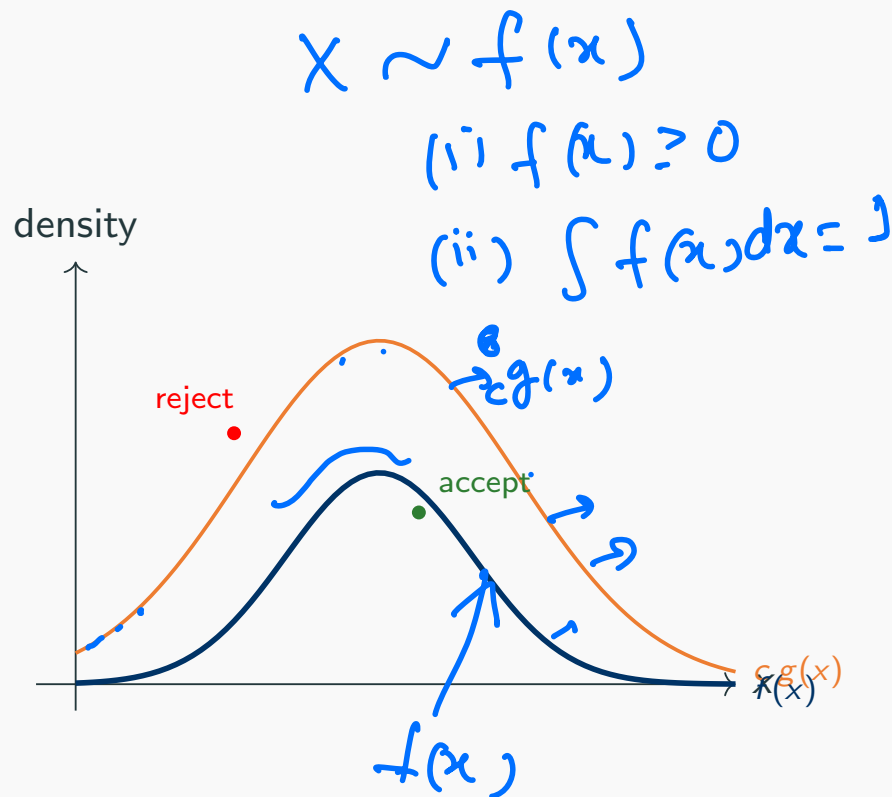


The Idea: Darts Under a Curve

- Target density $f(x)$, hard to sample directly.
- Choose an **envelope/proposal** density $g(x)$ that is easy to sample from.
- Choose a constant $c \geq 1$ such that

$$f(x) \leq c g(x) \quad \text{for all } x.$$

Handwritten notes: "target pdf" with an arrow pointing to $f(x)$, and an arrow pointing to $g(x)$.

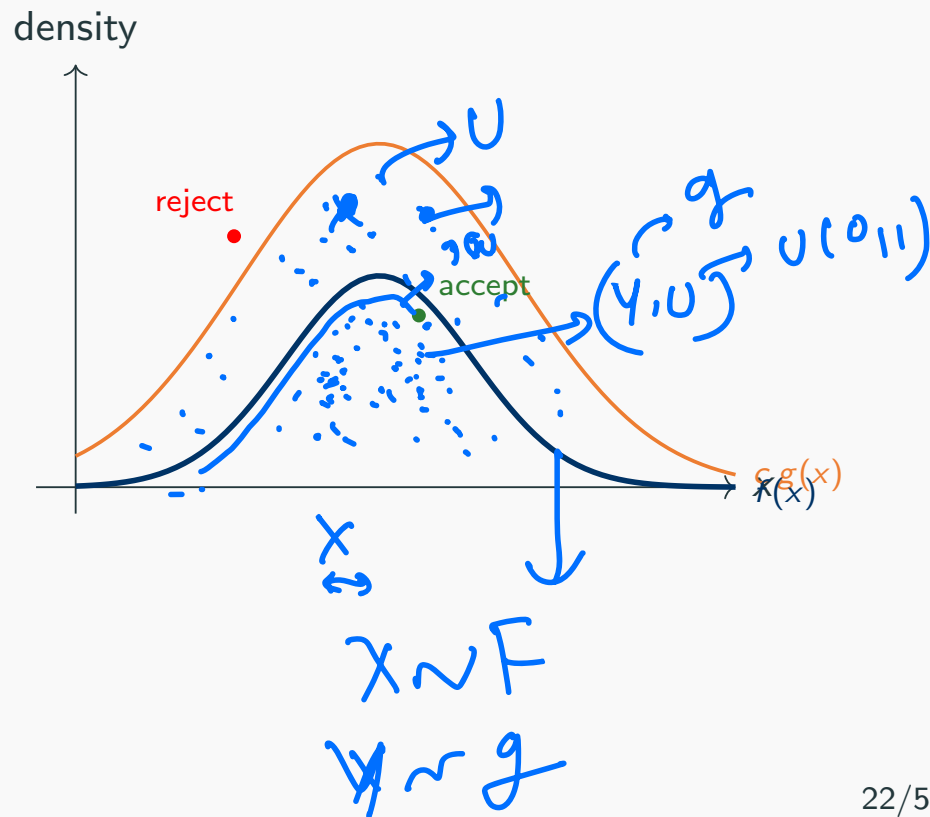


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$$f(x) \leq c g(x) \quad \text{for all } x.$$

- **Idea:** sample uniformly “under” the curve $c g(x)$; keep only the points that also fall under $f(x)$.



Formal Algorithm

Algorithm: Acceptance-Rejection Sampling

1. Generate $Y \sim g$.
2. Generate $U \sim \text{Uniform}(0, 1)$, independent of Y .
3. If

$$U \leq \left(\frac{f(Y)}{c g(Y)} \right)$$

accept $X = Y$. Otherwise reject and go to Step 1.

$$U \sim U(0, 1)$$

$$f(x) \leq c g(x)$$

$$c \geq \frac{f(x)}{g(x)} \quad \forall x$$

$$c = \sup_x \frac{f(x)}{g(x)}$$

$$U_3 \geq \frac{f(Y_3)}{c g(Y_3)}$$

$$U_4 \leq \frac{f(Y_4)}{c g(Y_4)}$$

$$\Rightarrow Y_1, Y_2, \dots, Y_n \sim g \rightarrow$$

$$U_1, U_2, \dots, U_n \sim U(0, 1)$$

$$U_1 \leq \frac{f(Y_1)}{c g(Y_1)}$$

$$U_2 \leq \frac{f(Y_2)}{c g(Y_2)}$$



$$X_1 = Y_1$$

$$X_2 = Y_2$$

$$X_3 = Y_4$$

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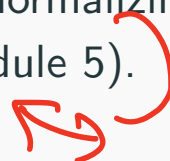
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$\rightarrow \frac{1}{c}$

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- Note c must satisfy $c = \sup_x \frac{f(x)}{g(x)}$ (or any valid upper bound) for the ratio in Step 3 to always lie in $[0, 1]$.
- Works even when f is known only up to a normalizing constant – a property that becomes crucial later, in Bayesian computation (Module 5).



Why Does This Work?

Proof sketch

Condition on acceptance. The joint “proposal” density of (Y, U) is $g(y)$ on y times uniform on $[0, 1]$ for u . The accepted region is $\{(y, u) : 0 \leq u \leq f(y)/(c g(y))\}$.

$$P(Y \leq x \mid \text{accept}) = \frac{\int_{-\infty}^x g(y) \cdot \frac{f(y)}{c g(y)} dy}{\int_{-\infty}^{\infty} g(y) \cdot \frac{f(y)}{c g(y)} dy} = \frac{\int_{-\infty}^x f(y) dy}{\int_{-\infty}^{\infty} f(y) dy} = F(x).$$

So the *accepted* Y values have exactly the target CDF F . ■

Efficiency of Acceptance–Rejection

- Each proposal Y is accepted with probability $P(\text{accept}) = \frac{1}{c}$.

$$\Rightarrow f(x, y) \\ f_x(x) = \int f(x, y) dy$$

$$P(\text{accept}) = \frac{1}{c}$$

$$P\left(\frac{U \leq \frac{f(Y)}{c g(Y)}}{c g(Y)}\right) = \int_{y=x} P\left(\frac{U \leq \frac{f(y)}{c g(y)}}{c g(y)}, \frac{y=x}{y=y}\right) dy$$

$$Y \sim g \\ U \sim U$$

$$= \int_{Y=y} P\left(U \leq \frac{f(y)}{c g(y)} \mid Y=y\right) g_Y(y) dy$$

$$P(A \cap B) = P(A|B) \cdot P(B)$$

$$= \int_{y=y}^y \underbrace{p\left(U \leq \frac{f(y)}{c g(y)} \mid Y=y\right)}_{\substack{U \sim U(0,1) \\ \frac{f(y)}{c g(y)}}} \underline{g(y)} dy$$

$$\int_0^{\frac{f(y)}{c g(y)}} 1 du = \frac{f(y)}{c g(y)}$$

$$= \int_{y=y} \frac{f(y)}{c g(y)} \cdot \cancel{g(y)} dy$$

$$= \int_{y=y} \frac{f(y)}{c} dy = \frac{1}{c} \underbrace{\int_{y=y} f(y) dy}_1$$

$$p(\text{acabl}) = \frac{1}{c}$$

Efficiency of Acceptance–Rejection

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Trade-off

A very tight envelope ($c \approx 1$) gives high efficiency but may be harder to construct or sample from; a loose, simple envelope is easy to sample but wastes many draws (small acceptance probability).

Example: Sampling from a Beta(2, 2) Density

Target: $f(x) = 6x(1-x)$, $0 \leq x \leq 1$.
Beta(2, 2)

Example: Sampling from a Beta(2, 2) Density

$$c g(x) \geq f(x)$$

Target: $f(x) = 6x(1-x)$, $0 \leq x \leq 1$.

Take **envelope** $g(x) = 1$ (i.e., Uniform(0, 1)), the simplest possible proposal.

$$c = \sup_{0 \leq x \leq 1} \frac{f(x)}{g(x)} = \sup_x 6x(1-x) = 6 \cdot \frac{1}{2} \cdot \frac{1}{2} = 1.5 \quad (\text{at } x = 0.5).$$

$$\frac{6x(1-x)}{1} \rightarrow \frac{d}{dx}(6x - 6x^2) = 0$$
$$\Rightarrow x = 0.5$$

$$\Rightarrow \frac{f(x)}{g(x)} = 6x(1-x) = 6 \times \frac{1}{2} \times \frac{1}{2} = 1.5$$

$$c = 1.5$$

$$\frac{f(x)}{c g(x)} = \frac{6x(1-x)}{1.5}$$

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Algorithm

1. Generate $Y \sim \text{Unif}(0, 1)$. ✓
2. Generate $U \sim \text{Unif}(0, 1)$. →
3. Accept $X = Y$ if $U \leq \frac{6Y(1 - Y)}{1.5} = 4Y(1 - Y)$; else repeat.

Expected number of proposals per acceptance = $c = 1.5$ – quite efficient.

$q = \text{Unif}(0, 1)$

→ $g(\cdot)$

$Y_1, Y_2, \dots, Y_n, Y_{n+1}, \dots$

$X_1 = Y_1, X_2 = Y_3, X_3 = Y_5, X_4 = Y_6$

$\dots X_n = Y_{n+1} \dots$

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$$g(x) = a e^{-a(x-a)}, \quad x \geq a,$$

chosen because it decays at a similar rate to the Normal tail.

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Lesson

The choice of envelope g is where the “art” of acceptance–rejection lies – match tail behaviour to the target for high efficiency.